

MANOJ SRIVATSAV

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EDUCATION

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Artificial Intelligence

Narayana Junior College

Telangana State Board of Intermediate Education

2023 – Present

CGPA: 7.5

2021 – 2023

Percentage: 94

RELEVANT COURSEWORK

Courses: Data Structures and Algorithms, Deep Learning, Cloud Computing, Machine Learning, Digital Forensics, Artificial Intelligence, Discrete Mathematics, Computer Networks, Mathematics for Computing, UI Design

SKILLS

Technical: Neural Networks, Python, Java, JavaScript, HTML/CSS, Regression Models

Frameworks: React, Node.js, Flask, JUnit, FastAPI, WordPress

Languages: English, Hindi, Telugu, Kannada

Soft Skills: Teamwork, Leadership, Attention to Detail, Competitive Analysis

PROJECTS

Emotional Detection Using Multi-Modal Analysis | *Python, OpenCV, Dlib, Librosa, TensorFlow*

- Built a multi-modal system combining facial expression and speech analysis for emotion detection.
- Implemented deep learning pipelines for FER and audio-based emotion recognition.
- Designed a real-time detection module demonstrating applications in HCI and mental health monitoring.

Quantum Maze Solver | *Qiskit, Python, Quantum Computing*

- Developed a quantum maze solver using superposition and Grover's algorithm for optimal pathfinding.
- Designed dynamic maze visualization to compare classical and quantum approaches.

Double-Pipe Heat Exchanger Performance Prediction | *Python, NumPy, Pandas, Keras, Scikit-Learn*

- Developed ML models (MLP, CNN, LSTM, GRU, BiLSTM) for prediction of friction factor, Nusselt number, and efficiency.
- Applied feature scaling, optimizer tuning, and MAE/MSE evaluation to select best models.
- Automated end-to-end pipelines for preprocessing, training, and generating Excel-based predictions.

Post-Earthquake Damage Assessment Using Siamese U-Net with Triplet Attention | *Python, PyTorch, U-Net, Attention*

- Built a damage-segmentation system using TE23D satellite data with pre/post-event imagery.
- Designed a Siamese U-Net with shared encoders and Triplet Attention for cross-dimensional feature refinement.
- Improved IoU, Dice, Precision, Recall, and F1 over U-Net, U-Net++, BIT, SegFormer, and BEiT baselines.

Green Video Streaming Optimization Framework | *Python, AWS EC2, Random Forest, Cloud Computing*

- Developed an energy-aware video transcoding system that optimizes **complexity, energy usage, and CO₂** emissions for cloud-based streaming.
- Implemented content-aware analysis and **Random Forest models** to predict CPU load, encoding time, and quality metrics for adaptive encoding.
- Deployed on **AWS EC2**, achieving up to **22.6% lower CPU usage**, **30% reduced energy consumption**, and **16.7% lower carbon emissions**.

INTERNSHIPS

Computer Vision Intern — FitFlix | *June 2024 – August 2024*

- Worked on a real-time **posture correction system** for fitness applications using Python, OpenCV, and pose-estimation landmarks.
- Implemented live angle computation for hips, knees, and ankles using custom functions (`find_angle`, `get_landmark_features`) and generated feedback for incorrect movements such as bending forward, knee-over-toe, and deep squats.
- Developed a full processing pipeline that tracked inactivity, counted correct/incorrect squats, displayed visual overlays, and produced real-time feedback using text overlays, dotted lines, and arc-based joint angle visualization.

POSITION OF RESPONSIBILITY

Fest Organizing Team | **2023 – Oct 2025**

- Core Member – Sponsorship & Marketing
- Core Member – Logistics Management